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Spatiotemporally prioritized experience replay for off-policy reinforcement learning

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Abstract

Off-policy reinforcement learning holds significant potential for solving complex sequential decision-making problems. However, how to improve its efficiency in learning from experiences is still a challenging issue for existing methods, especially in continuous action domains. To effectively utilize experiences, this paper proposes a new experience replay method, termed spatiotemporally prioritized experience replay (STP), which establishes both temporal and spatial priority measures for the experience. Concretely, the temporal priority of an experience is proportional to the time step when the experience was generated, while the spatial priority is designed to assign higher priority to the experience located in sparse regions. By leveraging these dual priorities, we can effectively mitigate the over-sampling of experiences generated at an early stage while simultaneously preventing the sampling bias towards dense regions. As a result, the sampled experiences possess both novelty and diversity, which are essential for efficient learning. Extensive experiments on several continuous action tasks confirm the superior performance of the proposed method.

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1. Introduction

Reinforcement learning (RL) is a paradigm for dealing with sequential decision-making problems [1]. It has demonstrated remarkable performance in many fields, such as games [2; 3], autonomous driving [4; 5], and natural language processing [6; 7]. At present, there are two main types of RL models: on-policy and off-policy [1]. Off-policy RL methods often achieve superior sample efficiency compared to on-policy methods by the experience replay mechanism, which enables historical experiences reuse [8]. However, most off-policy RL methods such as Deep Q-Network (DQN) [9] replay historical experiences uniformly at random, without mining the useful information contained in the experiences, undermining the potential benefits of experience reuse.

To enhance experience utilization, researchers have made substantial efforts on experience replay. Prioritized Experience Replay (PER) [10] assigns a priority score to important experiences, where the importance of an experience is measured by the Temporal Difference (TD) error. It has been shown to improve the performance of DQN. However, many empirical studies indicate that PER performs poorly when combined with actor-critic algorithms, such as Twin Delayed Deep Deterministic Policy Gradient (TD3) [11], which is widely used on continuous action tasks. As an optimized variant of PER, Loss-Adjusted Approximate Actor Prioritized experience replay (LA3P) [12] addresses PER’s shortcomings in actor-critic algorithms by prioritising low TD error experiences for actor training and high TD error experiences for critic training while incorporating random sampling for balance. It has demonstrated more promising results than PER on several continuous action tasks. Recently, Q-value-based Experience Replay (QER) [13] demonstrates that maintaining the diversity of sampled experiences can well enhance the efficiency of experience utilization. It hierarchically samples experiences based on the Q-values of their state-action pairs, which makes its performance sensitive to the accuracy of the Q-value estimator. In addition, Frequency and Similarity Experience Replay (FSER) [14] prioritizes the experiences based on frequency and similarity, where the experiences it samples consist of two parts: one part consists of experiences that are less frequently sampled, and the other part consists of experiences similar to the current policy. Although FSER alleviates unbalanced sampling, it still risks redundant sampling of experiences that are rarely visited yet closely similar to the current policy.

38 This paper proposes a new experience replay method, named SpatioTemporal
 39 Prioritized experience replay (STP), to prioritize recent experiences
 40 located in sparse regions. To this end, STP defines a spatiotemporal pri-
 41 ority for the experience by integrating newly defined temporal and spatial
 42 priorities, which are based on the temporal and spatial distribution of ex-
 43 periences, as illustrated in Fig. 1. In particular, for each experience, its
 44 temporal priority is proportional to the time step at which it was generated.
 45 By the temporal priority, we can effectively mitigate the over-sampling of
 46 experiences generated during the early stages of training. However, if one
 47 relies solely on the temporal priority and ignores the spatial distribution of
 48 experiences, the sampled experiences will typically be concentrated in dense
 49 regions. To address this, we introduce the spatial priority for an experience
 50 based on the density of the region where the experience is located, where ex-
 51 periences located in sparse regions are assigned higher priorities. Note that
 52 the state-action pair is the core component of an experience, as the remaining
 53 two elements (reward and next state) are contingent upon it. Accordingly,
 54 we quantify the spatial priority of each experience by evaluating the density
 55 of its corresponding state-action pair. By integrating these two newly defined
 56 priorities, we propose STP to encourage the sampling of the recently gen-
 57 erated experiences while suppressing redundant experience. To evaluate the
 58 effectiveness of STP, we integrate it with TD3 [11] and conduct experiments
 59 on five continuous action benchmarks, simulated by MuJoCo [15]. Empiri-
 60 cal results indicate that the proposed method achieves superior performance
 61 compared to baseline methods on most tested benchmarks.

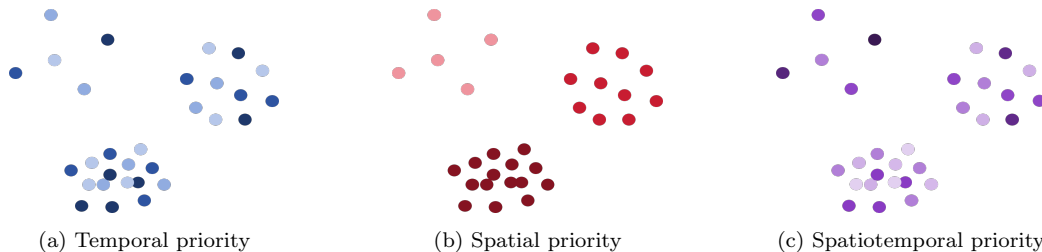


Figure 1: Visualization of the proposed priorities. In each figure, each point represents a distinct experience, and the distribution of the points corresponds to that of the state-action pairs of the experiences. Furthermore, the color intensity of each point reflects its priority level, where darker shades indicate higher priority.

62 The main contributions of this paper are summarized as follows:

- For each experience, we define a temporal priority to emphasize recently generated experiences and a spatial priority to avoid the sampling of redundant experiences within dense regions.
- Based on these spatial and temporal priorities, we propose a new experience replay method, STP, which prioritizes recently generated experiences located in sparse regions to enhance sample efficiency.
- We integrate the proposed method into the popular off-policy RL algorithm (TD3) and evaluate its performance on several continuous action tasks. The results demonstrate that our method performs better than other competing algorithms.

The remainder of this paper is organized as follows: Section 2 and Section 3 provide an overview of the background and related work, respectively. The proposed method is detailed in Section 4. Section 5 provides a series of experiments and shows the experimental results. Finally, Section 6 concludes this work and discusses potential future research directions.

2. Related work

Experience replay allows agents to remember and reuse historical experiences [8]. Uniform random replay is a classic experience replay method yet remains popular [9; 11; 16]. Although it can well break the temporal correlations of experiences for the updates, it fails to exploit the useful knowledge contained in the experiences, leading to suboptimal sample efficiency [17].

At present, most experience replay methods rely on the idea that the agent often benefits more by learning from some experiences than from others [10; 18; 19; 20]. PER [10], a well-known non-uniform experience replay method, prioritizes the experiences with high TD-errors. Inspired by PER, High-reward Experiences-assisted Policy Iteration (HEPI) [19] facilitates the frequent reuse of high-reward experiences, motivated by the human inclination to reflect on critical past events. Reward Prediction Errors (RPE) [21] replaces TD error in PER with the reward prediction error, which is the discrepancy between actual rewards and the predicted rewards generated by an enhanced model critic network. In addition, Liu et al. [22] prioritize experiences proximal to terminal states, utilizing a backtracking mechanism to propagate priority peaks throughout the training process. Instead of relying on specific metrics, some methods improve PER by combining TD-errors

97 with other criteria [23; 18; 20]. High-value PER [23] determines experi-
 98 ence priority by integrating TD errors with supplementary value metrics,
 99 including rewards and state-action values. Model-augmented PER [18] pri-
 100 oritizes experiences characterized by both large TD errors and high model
 101 estimation errors, utilizing these metrics to compute sampling probabilities
 102 and importance-sampling weights. Liu et al. [24] propose a multifaceted ap-
 103 proach that dynamically integrates four criteria (TD error, state entropy,
 104 reward and replay frequency) to comprehensively quantify the importance of
 105 each experience. Recently, Rewards-sensitive PER [20] has extended PER by
 106 incorporating a reward-sensitive priority mechanism that explicitly promotes
 107 transitions associated with above-average returns. It effectively modulates
 108 replay priorities by dynamically integrating TD errors with instantaneous
 109 reward signals, thereby ensuring that critical experiences are sampled with
 110 higher frequency. While prioritization of more useful samples is helpful, this
 111 strategy can also lead to overfitting, as useful samples are likely to be more
 112 rare [25].

113 To alleviate overfitting, some works increase the sampling frequency of
 114 recently gathered experiences [26; 27; 28], or high-rarity experiences [29; 30].
 115 Emphasizing Recent Experience (ERE) [31] demonstrates that under uni-
 116 form sampling, the sampling probability of older transitions disproportion-
 117 ately exceeds that of newer ones, particularly as the replay buffer nears its full
 118 capacity. It samples uniformly from the most recent $\eta^k N$ experiences for the
 119 k -th update, where N is the maximum buffer size and η is a parameter. Ex-
 120 perience Replay decay (ER decay) [32] also reduces the sampling probability
 121 of older experiences, which parallels infantile amnesia in its ability to prune
 122 the agent’s outdated experiences. On the other hand, Balanced Prioritized
 123 Experience Replay (BPER) [26] modulates the sampling frequency based on
 124 experiential rarity. It prioritizes experiences containing rare features while
 125 suppressing those with common patterns. In BPER, rarity is calculated by
 126 the occurrence probability of specific features within the state representation
 127 of a given experience. Based on three-way clustering [33], three-way experi-
 128 ence replay [28] enhances the sampling and eviction mechanisms to preserve
 129 the historical experiences with rare patterns. In addition, Xu et al. [30] in-
 130 troduce an adaptive approach to manage the buffer size by determining when
 131 to shrink the replay buffer and removing redundant samples automatically.
 132 Our work incorporates both the spatial sparsity and temporal recency of the
 133 experience, giving priority to recent and infrequent experiences.

134 In recent years, diversity-based experience replay methods have been

135 proven to enhance the efficiency of experience utilization. Dual-Priority
 136 Experience Replay (DPER) [34] utilizes a decoupled dual-priority architec-
 137 ture to categorize experiences into high- and low-priority streams based on a
 138 TD-error threshold, thereby alleviating redundancy caused by the prolonged
 139 storage of high-error, low-utility experiences. Efficient Diversity-based Expe-
 140 rience Replay (EDER) [35] employs a determinantal point process to model
 141 the diversity between samples and prioritizes replay based on the diversity
 142 between samples. When the vectors are mutually orthogonal, the volume
 143 and thus the determinant reaches its maximum value, reflecting the highest
 144 independence and diversity of the state representations. FSER [14] proposes
 145 a sampling method to prioritize the experiences that are rarely sampled or
 146 highly relevant to the current policy. Its sampling experience includes two
 147 parts, $(1 - \iota)b$ experiences based on frequency and samples ιb experiences
 148 based on similarity, where ι and b denote a parameter and the mini-batch
 149 size, respectively. Although FSER alleviates imbalanced sampling, the expe-
 150 riences it samples may contain overlapping experiences that are both rarely
 151 sampled and highly relevant to the current policy. To maintaining diversity
 152 within the sampled experiences, LA3P [12] adopts a hybrid sampling strat-
 153 egy that incorporates uniform, low-TD-error, and high-TD-error experiences.
 154 In addition, QER [13] promotes sampling diversity by utilizing a hierarchi-
 155 cal strategy that selects experiences based on the estimated Q-values of the
 156 state-action pairs. Instead of relying on dynamically updated metrics like
 157 TD errors or Q-values, this study focus on exploring the diversity of the
 158 spatiotemporal distribution of the sampled experiences.

159 **3. Background**

160 RL formalizes the interaction between an agent and its environment as
 161 a Markov decision process (MDP) [1]. The MDP is defined by the tuple
 162 $(\mathcal{S}, \mathcal{A}, \mathcal{R}, P)$, where $\mathcal{S}$ is the set of states, $\mathcal{A}$ is the set of actions, $\mathcal{R}$ is the
 163 set of rewards, and P is the environment dynamics model satisfying the
 164 Markov property. For off-policy RL, the agent first observes a state s_t of
 165 the environment, and then chooses an action $a_t \in \mathcal{A}$ by a behavior policy
 166 ψ . Finally, the agent takes the action to the environment and receives a
 167 reward r_{t+1} and a new state s_{t+1} according to the environment dynamics P .
 168 The tuple $(s_t, a_t, r_{t+1}, s_{t+1})$ can be stored in a replay buffer that is used to
 169 optimize the target policy π .

170 The Q-function (critic) serves as the standard mechanism for evaluating

171 the performance of the target policy. It describes the expected discounted
 172 cumulative reward following the execution of action a_t in state s_t :

$$Q(s_t, a_t) = \mathbb{E}_{r_{i>t}, s_{i>t} \sim P, a_{i>t} \sim \pi} \left[\sum_{i=t}^{\infty} \gamma^{(i-t)} r_{i+1} \mid s_t, a_t \right], \quad (1)$$

173 where $\gamma \in [0, 1)$ represents the discount factor that governs the trade-off
 174 between immediate and long-term rewards. The Q-function is determined
 175 by the Bellman equation [36]:

$$Q(s_t, a_t) = \mathbb{E}_{r_{t+1}, s_{t+1} \sim P} [r_{t+1} + \gamma \mathbb{E}_{a_{t+1} \sim \pi} [Q(s_{t+1}, a_{t+1})]]. \quad (2)$$

176 DQN [9] uses a deep neural network Q_θ to approximate the Q-function,
 177 which is trained by minimizing the loss $L_Q(\theta)$:

$$L_Q(\theta) = \mathbb{E}_{s_t \sim \rho^\psi, a_t \sim \psi, r_{t+1}, s_{t+1} \sim P} [(Q_\theta(s_t, a_t) - y_t)^2], \quad (3)$$

178 where

$$y_t = r_{t+1} + \gamma \max_{a^*} Q_{\theta'}(s_{t+1}, a^*), \quad (4)$$

179 ρ^ψ is the discounted state visitation distribution under the behavior policy ψ ,
 180 and $Q_{\theta'}$ is a separate target Q-network used to maintain stability in learning.
 181 Note that the parameters θ' are periodically updated by copying parameters
 182 of the Q-network Q_θ .

183 For continuous action tasks, the maximum in Eq. (4) to select the action
 184 a^* is intractable. To determine the action based on the state, actor-critic
 185 methods generally employ an actor network to parameterize the policy π_ϕ ,
 186 parameterized by ϕ . Below, we introduce a typical actor-critic algorithm,
 187 TD3 [11], which is the base algorithm used in our experiments.

188 TD3 [11] uses two critic networks to limit possible overestimation in actor-
 189 critic methods. For $i = 1, 2$, the critic networks Q_{θ_i} are updated by minimiz-
 190 ing the loss $L_Q(\theta_i)$:

$$L_Q(\theta_i) = \mathbb{E}_{s_t \sim \rho^\psi, a_t \sim \psi, r_{t+1}, s_{t+1} \sim P} [(Q_{\theta_i}(s_t, a_t) - y_t)^2], \quad (5)$$

191 where

$$y_t = r_{t+1} + \gamma \min_{j=1,2} Q_{\theta'_j}(s_{t+1}, \pi_{\phi'}(s_{t+1}) + \xi), \quad (6)$$

192 $\pi_{\phi'}$ and $Q_{\theta'_i}$ denote the target networks corresponding to the actor π_ϕ and
 193 critics Q_{θ_i} , respectively, ξ is clipped Gaussian noise. After the critic network

194 has been updated for a specified number of iterations, by the deterministic
 195 policy gradient method [37], the actor network π_ϕ is updated to minimize
 196 the loss $L_\pi(\phi)$:

$$L_\pi(\phi) = \mathbb{E}_{s_t \sim \rho^\psi} [-Q_{\theta_1}(s_t, a_t) |_{a_t = \pi_\phi(s_t)}]. \quad (7)$$

197 Note that, $\psi(s_t) = \pi_\phi(s_t) + \omega$, where ω is Gaussian noise. Additionally,
 198 the weights of the target networks can be updated with the soft update, that
 199 is, $\theta'_i \leftarrow \zeta \theta_i + (1 - \zeta) \theta'_i, i = 1, 2, \phi' \leftarrow \zeta \phi + (1 - \zeta) \phi'$, where $0 < \zeta \ll 1$.

200 4. Spatiotemporally prioritized experience replay

201 In this section, we first define the spatial priority of an experience based
 202 on the cardinality of the group where the experience is located, followed by
 203 the temporal priority of an experience determined by its generation time step.
 204 Subsequently, we propose the spatiotemporal priority by incorporating both
 205 spatial and temporal priorities. Finally, we introduce a replaying strategy
 206 based on the spatiotemporal priority.

207 4.1. Spatiotemporal priority

208 Let $\mathcal{B} = \{\beta_t\}_{t=1}^T$ be a replay buffer, where β_t represents the experience
 209 collected by the agent at each time step t . Generally, the replay buffer $\mathcal{B}$ is
 210 initialized through random exploration prior to training [11].

211 We first mine k groups $\{G_i\}_{i=1}^k$ in initial $\mathcal{B}$ according to the spatial dis-
 212 tribution of the state-action pairs of experiences. The group G_i is a set of
 213 experiences in a region centered at a representative experience g_i with a fixed
 214 radius of ϵ . We aim for that the state-action pairs of the representative expe-
 215 riences are uniformly distributed in the state-action space. Thus, we cluster
 216 the state-action pairs of the experiences in initial $\mathcal{B}$ into k clusters by k -
 217 means++ algorithm [38], and then take the experience whose state-action
 218 pair is closest to the cluster center as a representative experience. To reduce
 219 the overlap between different groups, we define the radius ϵ as the average
 220 of the minimum distances from the state-action pair of each representative
 221 experience to those of others:

$$\epsilon = \frac{1}{k} \sum_{g_i \in \mathcal{G}_0} \min_{g_j \in \mathcal{G}_0, i \neq j} d(x_i, x_j) \quad (8)$$

222 where $\mathcal{G}_0$ is the set of the k representative experiences, x_i is the vector of
 223 the state-action pair of β_i , and $d(x_i, x_j)$ is the Euclidean distance between

224 x_i and x_j . Note that if an experience lies within the ϵ -neighborhood of
 225 multiple representative experiences, we assign the experience to the group
 226 whose representative experience is closest to it.

227 In general, the experiences in $\mathcal{B}$ cannot be covered by k initial groups. For
 228 an experience $\beta_i \in \mathcal{B}$, if it is not covered by any existing group, then we set
 229 β_i as a representative experience of a new group. Formally, the experience
 230 β_i is identified as a new representative experience if it satisfies the following
 231 condition:

$$\beta_i \notin \bigcup_{g_j \in \mathcal{G}} G_j \quad (9)$$

232 where $\mathcal{G}$ denotes the set of all existing representative experiences.

233 We now define a larger spatial priority for the experience covered by
 234 sparse group while a smaller spatial priority for the experience covered by
 235 dense group. To be specific, for each experience β_i , its spatial priority u_i is
 236 defined as follows

$$u_i = e^{\text{clip}(v_i, 1-\alpha, 1+\alpha)} \quad (10)$$

237 where

$$v_i = \frac{\bar{\rho}}{|G_j|}, \quad (11)$$

238 $|X|$ denotes the cardinality of the set X , G_j is the group that contains β_i , $\bar{\rho}$
 239 is the average density of the group, $\bar{\rho} = |\mathcal{B}|/|\mathcal{G}|$, and α is a hyperparameter
 240 that satisfies $\alpha \in (0, 1)$. Note that the clipping operation ensures a mod-
 241 erate adjustment, avoiding overly large disparities across groups while still
 242 preserving the intended balance.

243 Next, we define a temporal priority to emphasize recently generated expe-
 244 riences. For the experience β_i generated during training, its temporal priority
 245 l_i is formally defined as $l_i = (i/m)^2$, where m denotes the capacity of the ini-
 246 tial replay buffer. Specifically, for any experience β_i within the initial replay
 247 buffer, let its temporal priority be $l_i = 1$.

248 Overall, the spatiotemporal priority of experience β_i , denoted as z_i , is
 249 defined as the sum of the weighted spatial priority and the temporal priority,

$$z_i = \lambda w_i u_i + (1 - \lambda) l_i, \quad (12)$$

250 where

$$w_i = \begin{cases} 1 & i \leq m, \\ \frac{1}{i-1} \sum_{j=1}^{i-1} z_j & \text{otherwise.} \end{cases} \quad (13)$$

Here, the weight w_i is the average spatiotemporal priority of existing experiences, and λ is a trade-off hyperparameter between these two types of priorities. Consequently, the experiences located in sparse regions and recently collected will be prioritized for sampling by the spatiotemporal priority.

4.2. *Replaying strategy*

At each training step, a mini-batch is sampled from the replay buffer using a non-uniform distribution defined by the spatiotemporal priorities. To be specific, the sampling probability of an experience β_i is defined as:

$$p_i = \frac{z_i}{\sum_j z_j}. \quad (14)$$

To efficiently sample the experiences, we employ a SumTree data structure [10; 12]. The SumTree is structurally analogous to a binary heap, with the distinction that each parent node stores the sum of its children’s values rather than adhering to the standard heap ordering property. Consequently, internal nodes represent intermediate sums, with the root node containing the total spatiotemporal priorities across all experiences, while leaf nodes store the individual spatiotemporal priorities. To sample a mini-batch of size b , we partition the range $[0, \hat{z}]$ into b intervals of equal width, where $\hat{z}$ denotes the total sum of the spatiotemporal priorities. Within each interval, a value is uniformly sampled.

To visually illustrate the superiority of STP over random experience replay, we combine both methods with TD3 and run experiments in the Ant environment for 1 million time steps. The results of the replay frequency per experience are shown in Fig. 2, with detailed statistics provided in Table 1. The empirical results confirm that our method yields greater sample diversity than random sampling, which is beneficial for sample efficiency. Additionally, we present a comparison of the dimensional variances for states and actions in the two replay buffers at the end of training in Fig. 3. As illustrated in Fig. 3, integrating our method with TD3 enables the agent to perform more diverse actions, which consequently leads to the exploration of more diverse states. This suggests that training on the novel and diverse experiences sampled by our method facilitates more effective exploration.

Table 1: Statistics of replay frequency across sampling methods.

	Max.	Mean	Std. Dev.
Random sampling	1081	249.4	111.6
STP	545	249.4	231.4

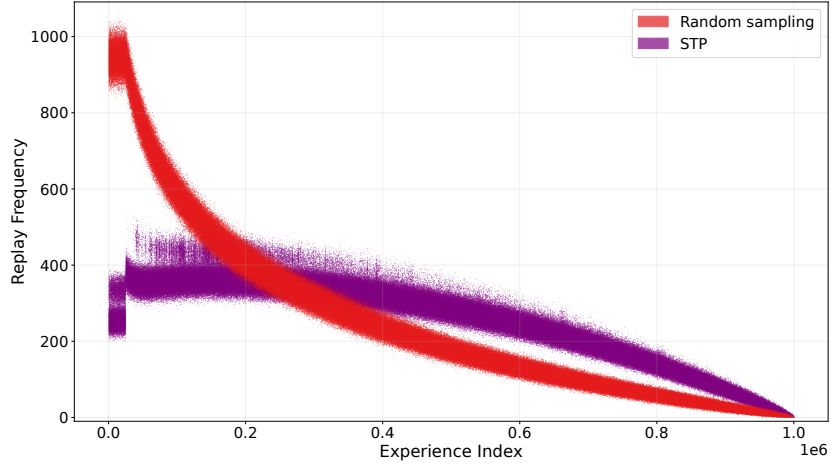


Figure 2: Comparison of the replay frequency between random sampling and STP.

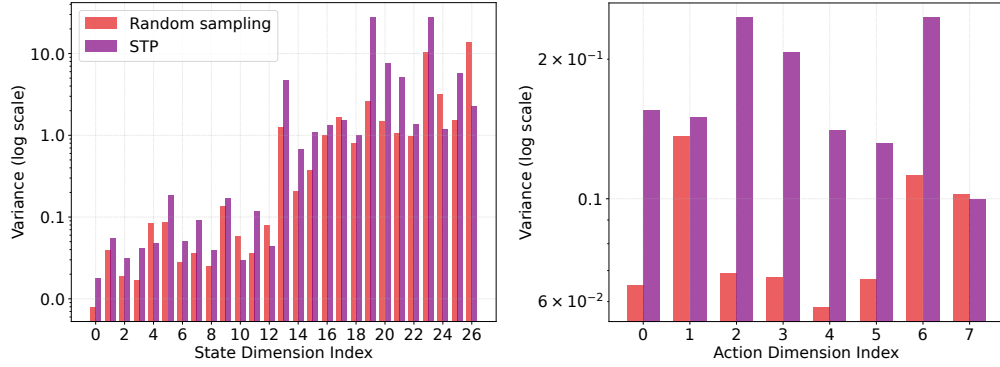


Figure 3: Variance comparison of state and action dimensions within the replay buffer.

281 5. Experiments

282 In this section, extensive experiments are conducted on five environments
 283 to demonstrate the superiority of STP. Specifically, we first describe the ex-
 284 perimental setup used to produce the reported results. We then compare
 285 the proposed method with baseline methods and perform a series of abla-
 286 tion studies to better understand STP. In addition, a sensitivity analysis is
 287 performed to evaluate the impact of λ , which dictates the trade-off within
 288 our method. To improve reproducibility, the source code of our method
 289 implementation is available online¹.

290 5.1. Experimental Setup

291 We evaluate the performance of our method on standard benchmarks
 292 from MuJoCo [15], accessed through the OpenAI Gym interface [39]. The
 293 environments used in the experiments are the v4 versions of Ant, HalfChee-
 294 tah, Hopper, Humanoid, and Walker2d, all of which are continuous action
 295 tasks. They cover a wide spectrum of control difficulties, ranging from low-
 296 dimensional planar locomotion tasks to high-dimensional whole-body coordi-
 297 nation problems. Details of these environments are summarized in Table
 298 2, and their visualizations are provided in Fig. 4.

Table 2: Statistics of the environment.

Algorithm	Action Dim.	State Dim.
Ant	8	27
HalfCheetah	6	17
Hopper	3	11
Humanoid	17	376
Walker2d	6	17

299 TD3 ² [11] is employed as our base algorithm, which utilizes a random
 300 experience replay method. The comparative methods include the well-known

¹<https://github.com/Royblue2/STP>

²<https://github.com/sfujim/TD3>

301 PER³ [10], and three recent methods, LA3P⁴ [12], RPE⁵ [21] and Blended Ex-
 302 ploitation and Exploration Actor-Critic (BAC)⁶ [40], all of which are imple-
 303 mented based on TD3. For fairness, all compared algorithms share common
 304 hyperparameters. Additionally, specific hyperparameters unique to each al-
 305 gorithm are set to their standard configurations as suggested by the authors.
 306 Table 3 summarizes the hyperparameters used in our experiments.

307 In our experiments, each algorithm is trained for one million environment
 308 steps, with evaluation conducted every 5000 time steps, where each evalu-
 309 ation reports the mean return over 10 deterministic test episodes without
 310 exploration noise. To account for stochasticity, each experiment is repeated
 311 with 10 random seeds, and results are reported as the mean return. In addi-
 312 tion, all algorithms are run on a machine with eight CPU cores (Intel Core
 313 i7-7820X 3.60GHz), 32 GB of memory and a single GeForce RTX 3070 GPU.

³<https://github.com/sfujim/LAP-PAL/tree/master/continuous>

⁴<https://github.com/baturaysaglam/LA3P>

⁵<https://github.com/UoA-CARES/RPE-PER>

⁶<https://jity16.github.io/BEE>

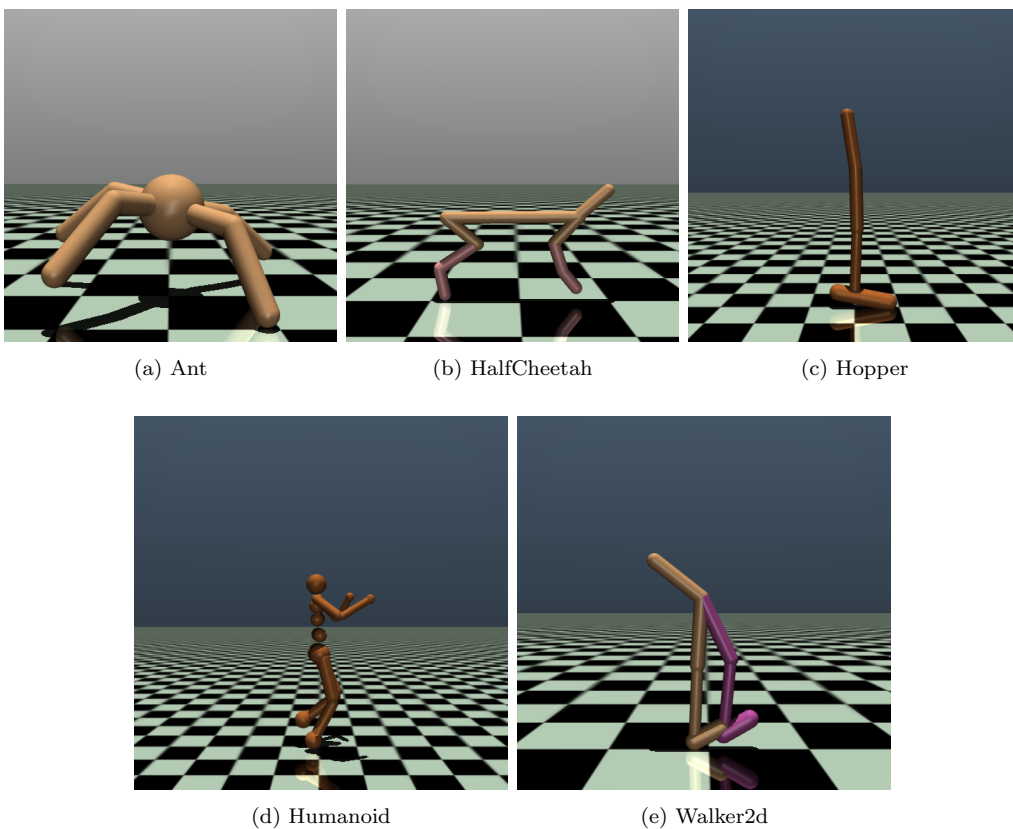


Figure 4: Visualizations of the environments.

Table 3: Hyperparameters used in the experiment.

Algorithm	Hyperparameter	Value
TD3	Replay buffer size m	1e6
	Maximum time steps T	1e6
	Evaluation interval	5000
	Initial random policy steps	25e3
	Discount factor γ	0.99
	Network	3-layer MLPs
	Number of hidden layer neurons	256
	Activation for hidden layers	ReLU
	Activation for output layers of actor	Tanh
	Activation for output layers of critic	None
	Optimizer	Adam
	Learning rate	3e-4
	Mini-batch size b	256
	Target policy noise clipping c	0.5
	Target network update rate τ	0.005
	Policy update frequency d	2
	Exploration noise σ_1	0.1
	Target policy noise σ_2	0.2
PER	Priority exponent α	0.6
	Importance sampling exponent β	0.4
LA3P	Uniform sampling fraction λ	0.5
	Priority exponent α	0.4
	Importance sampling exponent β	0.4
RPE	Priority exponent α	0.7
	Importance sampling exponent β	0.4
	Minimum priority	1.0
BAC	Quantile τ	0.7
	Balance factor λ	0.5
Ours	Number of initial groups k	10
	Spatial priority clipping α	0.2
	Trade-off factor λ	0.5

314 5.2. Performance Evaluation

315 Learning curves for benchmarks are reported in Fig. 5. The solid line
 316 indicates the mean performance, while the shaded region indicates the cor-
 317 responding half-standard deviation derived from 10 independent runs. As
 318 shown in Fig. 5, STP outperforms the baselines in almost all tested environ-
 319 ments. Specifically, STP achieves significant improvements over the baseline
 320 algorithms in environments HalfCheetah and Walker2d, where the Walker2d
 321 environment is unstable according to the analysis by Fujimoto et al. [41].
 322 In addition, we also observe a notable improvement of our method over pre-
 323 vious method in the Humanoid environment, demonstrating its ability to
 324 high-dimensional action and state spaces. While BAC and LA3P show com-
 325 petitive performance in the Ant and Hopper environments, respectively, their
 326 performance are inferior to that of STP in the other environments.

327 The quantitative results of the final evaluation for each algorithm are pre-
 328 sented in Table 4. Based on the data listed in Table 4, STP achieves superior
 329 performance in terms of both mean and maximum returns across all environ-
 330 ments except for the Hopper environment. Although LA3P achieves the high-
 331 est average return in the Hopper environment, our method also demonstrates
 332 competitive performance. Note that, in the HalfCheetah and Walker2d envi-
 333 ronments, STP offers a significant advantage with average returns of 11636.9
 334 and 4630.0, clearly outperforming the second-best baselines, BAC (10599.4)
 335 and LA3P (4152.1), respectively. The experimental results validate the ca-
 336 pability of STP to enhance sample efficiency compared to baseline methods.

337 As illustrated in Fig. 6, we evaluate the run time performance of STP
 338 against the baselines throughout 1 million training time steps in the Ant en-
 339 vironments. The execution time of LA3P and BAC in the ant environment
 340 is approximately twice that of the base algorithm, and is also significantly
 341 higher than that of the proposed method. Although the execution time of
 342 STP is slightly higher than those of PER and RPE, it significantly outper-
 343 forms both algorithms across all test environments. The results indicate
 344 that our method favorable trade-offs between computational cost and per-
 345 formance.

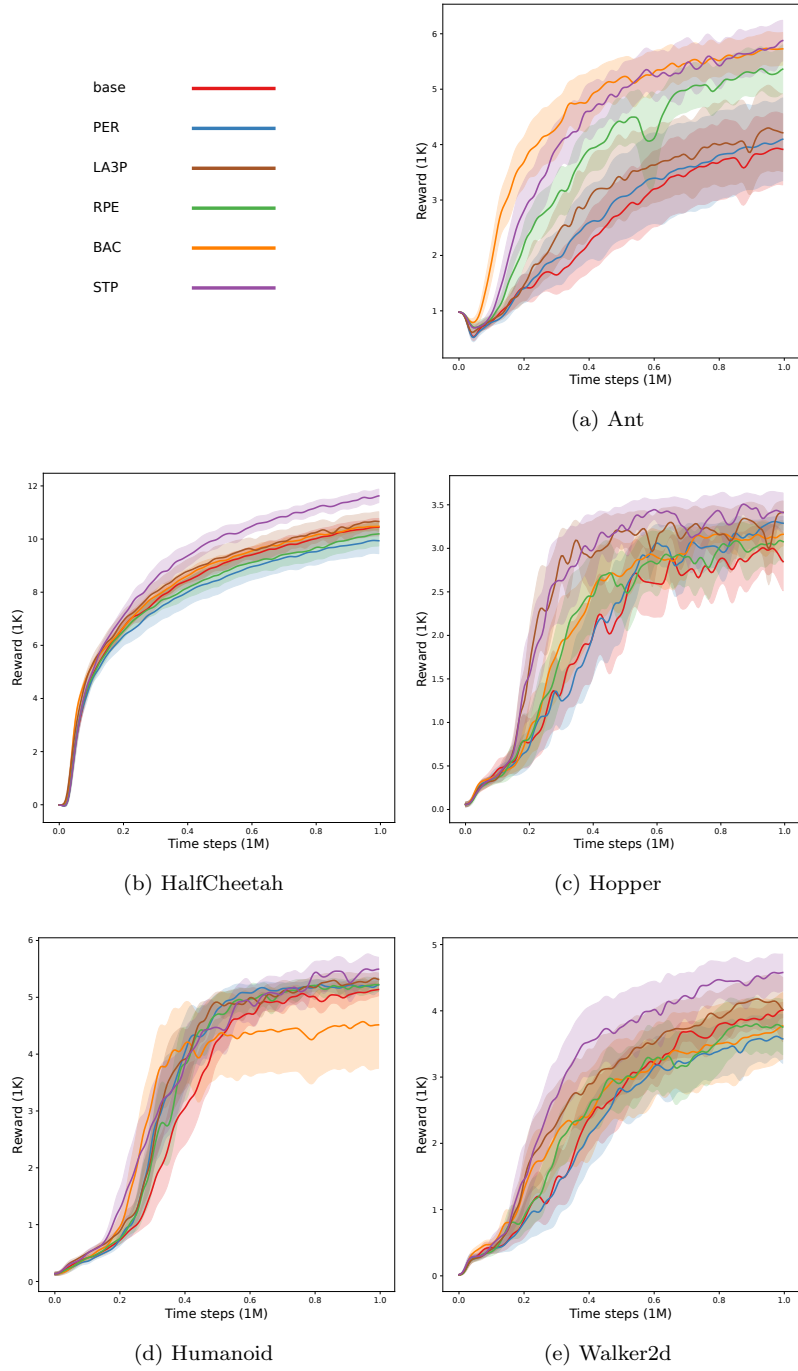


Figure 5: Learning curves of STP and the baselines.

Table 4: Comparison results of STP and the baselines.

Environment		base	PER	LA3P	RPE	BAC	STP
Ant	mean	3922.7	4052.1	4180.8	5355.3	5752.6	5890.6
	max	5579.9	5616.7	5927.2	6095.8	6139.5	6432.1
HalfCheetah	mean	10439.1	9962.5	10547.6	10256.2	10599.4	11636.9
	max	11097.3	10914.3	11634.8	11507.8	10966.4	12424.5
Hopper	mean	2712.2	3243.4	3321.7	3012.2	3157.5	3320.9
	max	3559.6	3498.0	3647.0	3456.2	3429.1	3649.4
Humanoid	mean	5014.5	5187.9	5379.5	5249.8	4593.1	5404.4
	max	5332.2	5498.0	5600.4	5679.5	5454.2	5927.7
Walker2d	mean	4054.7	3508.9	4152.1	3644.0	3766.4	4630.0
	max	4471.4	4410.5	5220.1	5233.9	5117.8	5541.5

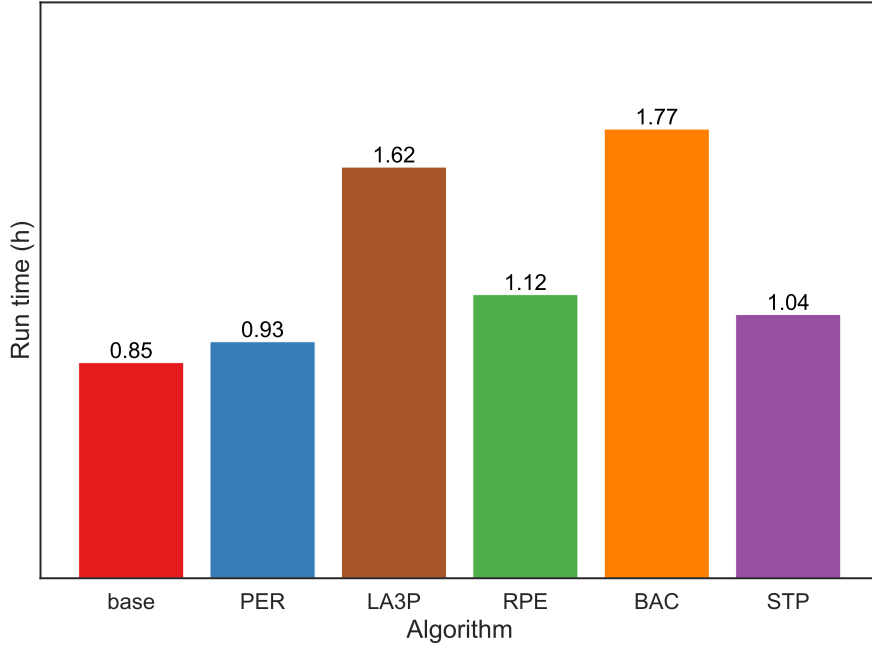


Figure 6: Run time of STP and the baselines in the Ant environment.

346 5.3. Ablation study

347 The STP algorithm introduces two designs to improve sample efficiency:
348 spatial priority (SP) and temporal priority (TP). To evaluate the individual
349 contribution of each component in STP, we first provide ablation studies in
350 this section. Then, we conduct the hyperparameter analysis on λ used to
351 trade off between temporal and spatial priorities, which is the key hyperpa-
352 rameter for our method.

353 Table 5 lists the quantitative results of the ablation study, while the learn-
354 ing curves are visualized in Fig. 7. Based on the results, comparing TP with
355 the base method reveals the significant contribution of prioritizing the re-
356 cently generated experiences. Similarly, SP shows consistent improvements
357 over the base method, proving the effectiveness of suppressing redundant
358 experience. As expected, STP consistently achieves the highest mean re-
359 turn in four out of the five environments (Ant, HalfCheetah, Humanoid, and
360 Walker2d), confirming the additive benefits of integrating both spatial and
361 temporal priorities. Specifically, in the Humanoid environment, STP achieves
362 a mean score of 5404.4, significantly outperforming the baseline (5014.5) and
363 surpassing both SP (5135.8) and TP (5331.7), where the Humanoid environ-
364 ment is widely recognized as one of the most challenging benchmarks in the
365 MuJoCo suite [41; 12]. Therefore, by the combination of SP and TP, STP
366 prioritizes newly generated experiences while limiting redundant experiences,
367 which helps to further enhance the efficiency of experience utilization.

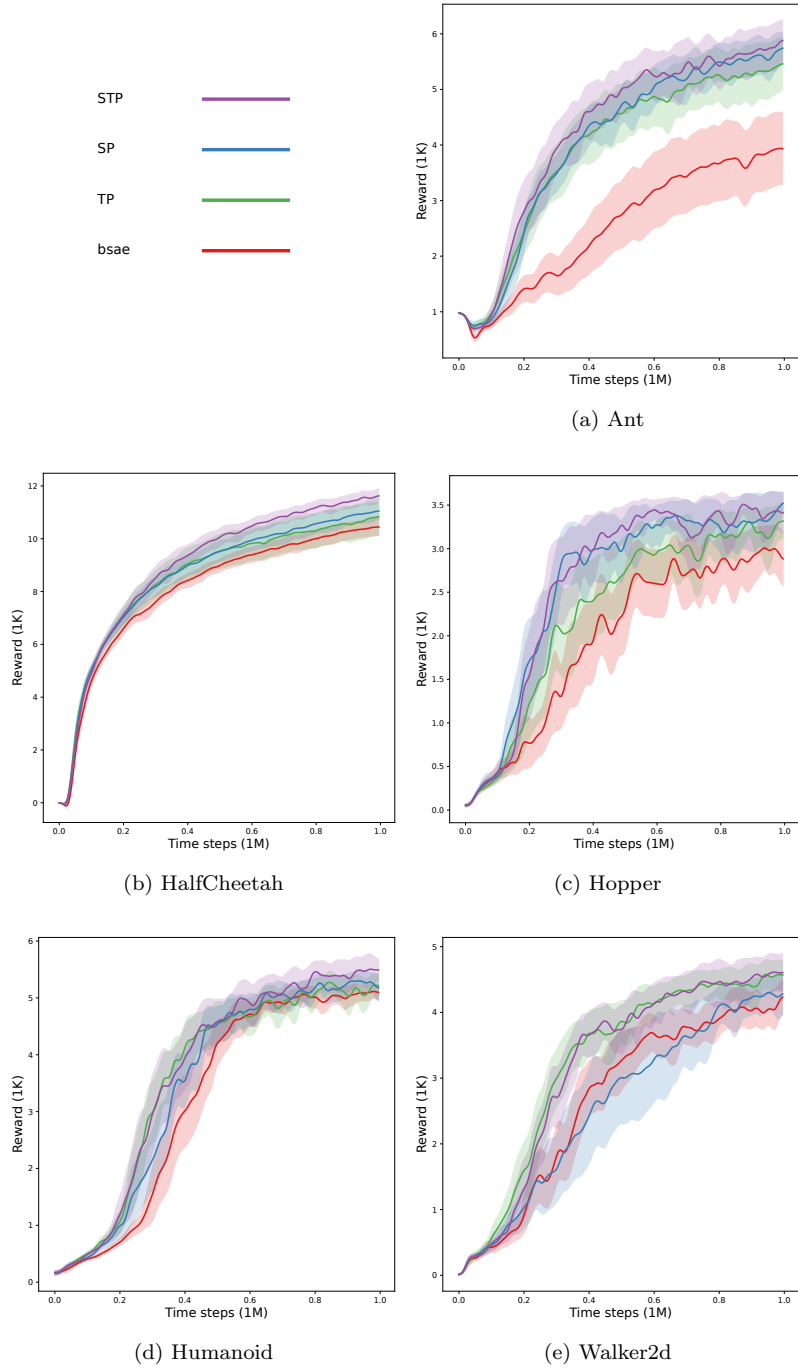


Figure 7: Learning curves for STP ablation study.

Table 5: Ablation results of STP.

		Ant	HalfCheetah	Hopper	Humanoid	Walker2d
STP	mean	5890.6	11636.9	3320.9	5404.4	4630.0
	max	6432.1	12424.5	3649.4	5927.7	5541.5
SP	mean	5838.1	11019.3	3563.5	5135.8	4354.3
	max	6202.0	11923.3	3749.8	5572.0	5215.0
TP	mean	5476.8	10843.7	3419.3	5331.7	4506.3
	max	6338.8	12089.9	3605.6	5715.0	5356.8
base	mean	3922.7	10439.1	2712.2	5014.5	4054.7
	max	5579.9	11097.3	3559.6	5332.2	4471.4

368 The learning curves for different values of hyperparameter λ are shown
 369 in Fig. 8, and the corresponding performance of the final policies is listed
 370 in Table 6. We evaluate λ across the set $\{0.1, 0.3, 0.5, 0.7, 0.9\}$. Accord-
 371 ing to the learning plots in Fig. 8, we observe that the proposed method
 372 achieves competitive performance across all selected parameters in all tested
 373 environments. This demonstrates that STP is not overly sensitive to the
 374 hyperparameter λ . In addition, $\lambda = 0.5$ produces the best mean returns in
 375 four out of the five test environments, as observed in Table 6. Consequently,
 376 choosing a moderate value for λ effectively balances spatial and temporal
 377 priorities, which enhances the robustness of the proposed method.

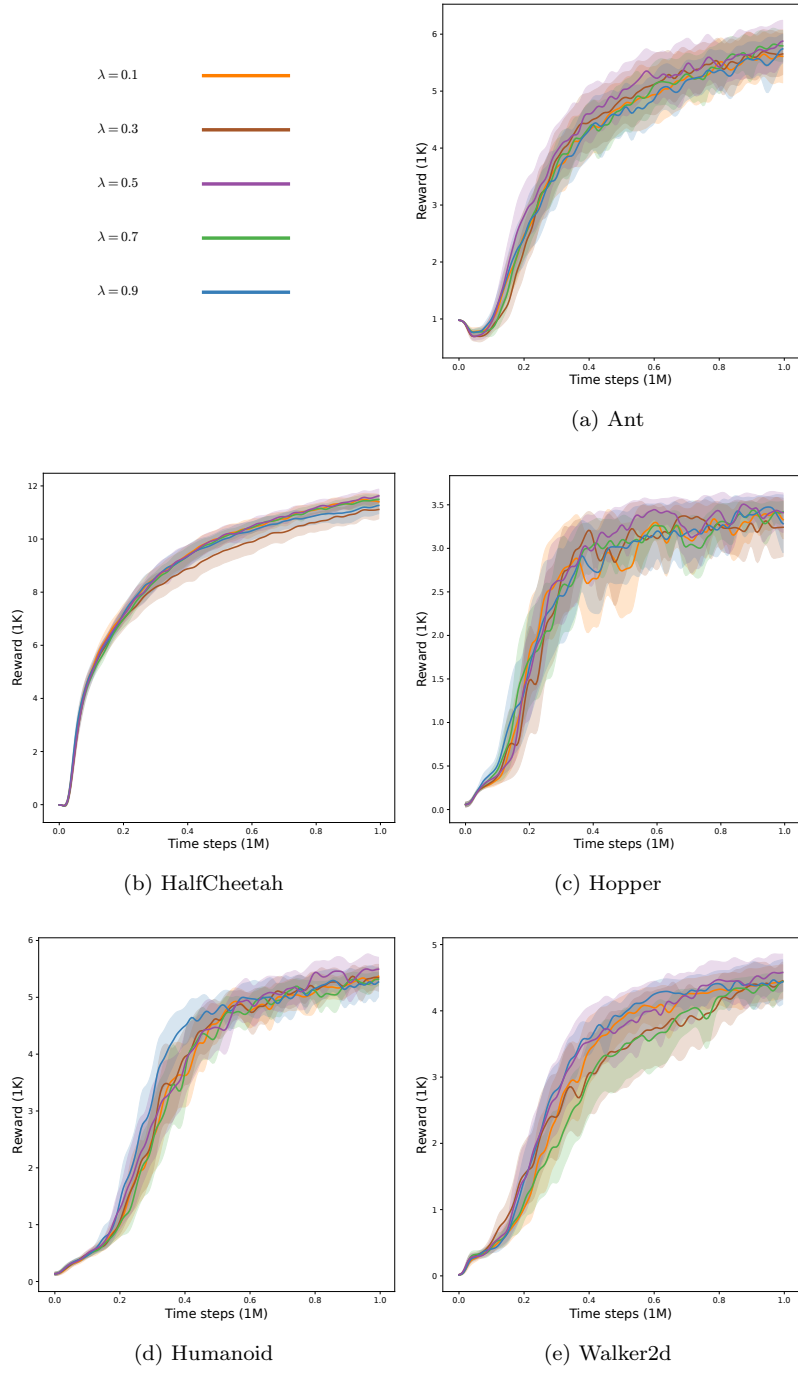


Figure 8: Learning curves for varying values of λ .

Table 6: Final policy performance for varying values of λ .

		Ant	HalfCheetah	Hopper	Humanoid	Walker2d
$\lambda = 0.1$	mean	5531.6	11270.1	3314.7	5373.0	4456.1
	max	6576.2	12064.5	3652.5	5923.9	5288.9
$\lambda = 0.3$	mean	5741.1	11193.5	3221.5	5305.6	4515.1
	max	6288.2	12388.8	3635.5	5975.1	5751.7
$\lambda = 0.5$	mean	5890.6	11636.9	3320.9	5404.4	4630.0
	max	6432.1	12424.5	3649.4	5927.7	5541.5
$\lambda = 0.7$	mean	5811.4	11479.3	3498.2	5309.2	4493.9
	max	6483.8	12078.5	3664.1	5664.3	4854.1
$\lambda = 0.9$	mean	5774.4	11294.6	3244.6	5377.2	4443.2
	max	6296.3	12038.4	3673.3	5983.7	5303.0

6. Conclusion

In this paper, we propose STP, an experience replay method designed to improve the learning efficiency of off-policy RL agents. STP encourages sampling of novel experiences while suppressing redundant ones. To this end, for each experience, its spatiotemporal priority is defined by integrating our defined temporal and spatial priorities, which are calculated based on the time of the experience generation and the density of the region where the experience is located, respectively. To verify that the proposed method is beneficial for improving utilization efficiency, we combine it with the well-known TD3 algorithm and construct experiments. Experiments on five MuJoCo environments clearly demonstrate that STP is well-suited for continuous action tasks. In the future, we will focus on extending STP to multi-agent RL settings [42; 43], aiming to mitigate the high degree of experience redundancy inherent in multi-agent interaction spaces.

CRediT authorship contribution statement

Wenyu Li: Methodology, Software, Data curation, Visualization, Validation, Writing - original draft. **Ruijia Li:** Conceptualization, Methodology, Software, Validation, Writing - original draft, Writing - review & editing, Supervision.

397 **Declaration of competing interest**

398 The authors declare that they have no known competing financial inter-
399 ests or personal relationships that could have appeared to influence the work
400 reported in this paper.

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Declaration of interests

☒The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

☐The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: